DHS 2026 H2 Math Preliminary Examination Paper 1 (Suggested Solutions)

Qn	Suggested Solutions
1	Let $\mathrm{f}(x) = Ax^3 + Bx^2 + Cx + D$ Using vertical asymptote at $x = -2$, when $x = -2$, "denominator = 0", i.e. $\mathrm{f}(-2) = 0$ $\Rightarrow A(-2)^3 + B(-2)^2 + C(-2) + D = 0$ $\Rightarrow -8A + 4B - 2C + D = 0 \ldots\ldots(1)$ Using Remainder Theorem, $\Rightarrow \mathrm{f}(2) = 4 \Rightarrow A(2)^3 + B(2)^2 + C(2) + D = 4$ $8A + 4B + 2C + D = 4 \ldots\ldots(2)$ Given $\left(-1, \frac{1}{4}\right)$ is a stationary point on the curve, $(-1, 4)$ is also a stationary point of f $\Rightarrow \mathrm{f}'(-1) = 0$ $\Rightarrow \mathrm{f}'(x) = 3Ax^2 + 2Bx + C$ $\Rightarrow \mathrm{f}'(-1) = 3A(-1)^2 + 2B(-1) + C = 0$ $\Rightarrow 3A - 2B + C = 0 \ldots\ldots(3)$ Given $\left(-1, \frac{1}{4}\right)$ is a point on the curve, when $x = -1, y = \frac{1}{4}$. $\Rightarrow \frac{1}{4} = \frac{1}{\mathrm{f}(-1)} \Rightarrow \mathrm{f}(-1) = 4$ $\Rightarrow A(-1)^3 + B(-1)^2 + C(-1) + D = 4$ $-A + B - C + D = 4 \ldots\ldots(4)$ Using GC to solve (1)-(4), $A = 1, B = 0, C = -3, D = 2$ Therefore, equation of curve is $y = \frac{1}{x^3 - 3x + 2}$ **<u>Alternative Method 1</u>**: Since $x = -2$ is a vertical asymptote, $(x + 2)$ is a factor of f(x) $\Rightarrow \mathrm{f}(x) = (x + 2)(ax^2 + bx + c)$ Using Remainder Theorem, $\Rightarrow \mathrm{f}(2) = 4 \Rightarrow (2 + 2)(a(2)^2 + b(2) + c) = 4$ $4a + 2b + c = 1 \ldots\ldots(1)$ Given $\left(-1, \frac{1}{4}\right)$ is a stationary point on the curve, $(-1, 4)$ is also a stationary point of f $\Rightarrow \mathrm{f}'(-1) = 0$

$\Rightarrow \mathrm{f}'(x) = (x+2)(2ax+b) + (ax^2+bx+c)$

$= 3ax^2 + (4a+2b)x + 2b + c$

$\Rightarrow \mathrm{f}'(-1) = 3a(-1)^2 + (4a+2b)(-1) + 2b + c = 0$

$\Rightarrow -a + c = 0 \ldots\ldots(2)$

Given $\left(-1, \frac{1}{4}\right)$ is a point on the curve, when $x = -1, y = \frac{1}{4}$.

$\Rightarrow \frac{1}{4} = \frac{1}{\mathrm{f}(-1)} \Rightarrow \mathrm{f}(-1) = 4 \Rightarrow (-1+2)(a(-1)^2 + b(-1) + c) = 4$

$a - b + c = 4 \ldots\ldots(3)$

Using GC to solve (1)-(3), $a = 1, b = -2, c = 1$

Therefore, equation of curve is $y = \frac{1}{(x+2)(x^2-2x+1)} = \frac{1}{x^3-3x+2}$.

Alternative Method 2:

Using Remainder Factor Theorem, since dividing by $(x-2)$ gives a remainder of 4

$\Rightarrow \mathrm{f}(x) = (x-2)(\alpha x^2 + \beta x + \gamma) + 4$

Since $x = -2$ is a vertical asymptote, $\mathrm{f}(-2) = 0$

$\Rightarrow \mathrm{f}(-2) = (-2-2)(4\alpha - 2\beta + \gamma) + 4 = 0$

$-16\alpha + 8\beta - 4\gamma = -4 \ldots\ldots(1)$

Given $\left(-1, \frac{1}{4}\right)$ is a stationary point on the curve, $(-1, 4)$ is also a stationary point of f $\Rightarrow \mathrm{f}'(-1) = 0$

$\Rightarrow \mathrm{f}'(x) = (x-2)(2\alpha x + \beta) + (\alpha x^2 + \beta x + \gamma)$

$= 3\alpha x^2 + (2\beta - 4\alpha)x + \gamma - 2\beta$

$\Rightarrow \mathrm{f}'(-1) = (-1-2)(2\alpha(-1) + \beta) + (\alpha(-1)^2 + \beta(-1) + \gamma) = 0$

$\Rightarrow 7\alpha - 4\beta + \gamma = 0 \ldots\ldots(2)$

Given $\left(-1, \frac{1}{4}\right)$ is a point on the curve, when $x = -1, y = \frac{1}{4}$.

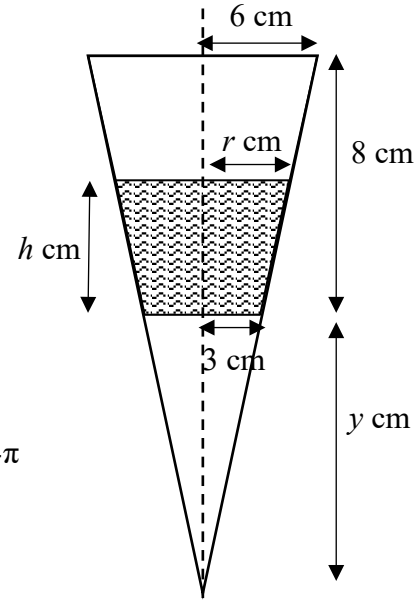
$\Rightarrow \frac{1}{4} = \frac{1}{\mathrm{f}(-1)} \Rightarrow \mathrm{f}(-1) = 4$

$\Rightarrow \mathrm{f}(-1) = (-1-2)(\alpha - \beta + \gamma) + 4 = 4$

$-3\alpha + 3\beta - 3\gamma = 0 \ldots\ldots(3)$

Using GC to solve (1)-(3), $\alpha = 1, \beta = 2, \gamma = 1$

Therefore, equation of curve is $y = \frac{1}{(x-2)(x^2+2x+1)+4} = \frac{1}{x^3-3x+2}$.

Qn	Suggested Solutions
2(a)	Let h and r be the height and radius of the water respectively. Using similar triangles, $\frac{y}{3} = \frac{y+8}{6} \Rightarrow y = 8$ Using similar triangles, $\frac{h+8}{r} = \frac{8}{3} \Rightarrow r = \frac{3}{8}(h+8)$ Volume of water, $V = \frac{1}{3}\pi r^2 (h+8) - \frac{1}{3}\pi(3^2)(8)$ $= \frac{1}{3}\pi\left(\frac{3}{8}(h+8)\right)^2 (h+8) - 24\pi$ $= \frac{3\pi}{64}(h+8)^3 - 24\pi$ (shown) 
2(b)	Differentiate with respect to t: $\frac{dV}{dt} = \frac{9\pi}{64}(h+8)^2 \times \frac{dh}{dt}$ $-15 = \frac{9\pi}{64}(2+8)^2 \times \frac{dh}{dt}$ $\frac{dh}{dt} = \frac{-16}{15\pi}$ $\therefore$ The height is falling at a rate of $\frac{16}{15\pi}$ cm s^{-1}.

Qn	Suggested Solutions
3(a)	$u_{n+1} > u_n$ for $n > 0$ $2u_n + p > u_n$ $-u_n < p$ for $n > 0$ Since $-u_{n+1} < -u_n$ for $n > 0$, $\ldots < -u_{n+1} < -u_n < \ldots < -u_1 = -1 < p$ $\therefore p > -1$ (shown)
3(b)(i)	$v_3 = 2a + 3$ $v_4 = 2b + 3$ $v_4 = 3v_3$ $2b + 3 = 6a + 9$ $2b = 6a + 6$ $b = 3a + 3$
3(b)(ii)	$v_6 = 2(2b + 3) + 3$ $= 4b + 9$ $= 12a + 21$

Qn	Suggested Solutions
4(a)	$\int \frac{\ln x}{x^2}\,\mathrm{d}x$ $= \ln x\left(-\frac{1}{x}\right) - \int\left(-\frac{1}{x}\right)\left(\frac{1}{x}\right)\mathrm{d}x$ $= -\frac{\ln x}{x} + \int \frac{1}{x^2}\,\mathrm{d}x$ $= -\frac{\ln x}{x} - \frac{1}{x} + A,$ where A is an arbitrary constant $u = \ln x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x}$ $\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{1}{x^2} = x^{-2}$ $v = -x^{-1} = -\frac{1}{x}$
4(b)	$12\int_3^4 \frac{\ln x}{x^2}\,\mathrm{d}x$ $= 12\left[-\frac{\ln x}{x} - \frac{1}{x}\right]_3^4$ (from part **(a)**) $= 12\left[-\frac{\ln 4}{4} - \frac{1}{4} - \left(-\frac{\ln 3}{3} - \frac{1}{3}\right)\right]$ $= 1 - 3\ln 4 + 4\ln 3$ $= 1 - 6\ln 2 + 4\ln 3$ $2\int_3^4 \frac{x}{(x-1)(x-2)}\,\mathrm{d}x$ $= 2\int_3^4 \left(\frac{2}{x-2} - \frac{1}{x-1}\right)\mathrm{d}x$ $= 2\left[2\ln|x-2| - \ln|x-1|\right]_3^4$ $= 2\left[2\ln 2 - \ln 3 - (2\ln 1 - \ln 2)\right]$ $= 6\ln 2 - 2\ln 3$ OR $\int_3^4 \frac{2x}{x^2 - 3x + 2}\,\mathrm{d}x$ $= \int_3^4 \frac{2x-3}{x^2-3x+2}\,\mathrm{d}x + \int_3^4 \frac{3}{\left(x-\frac{3}{2}\right)^2 - \left(\frac{1}{2}\right)^2}\,\mathrm{d}x$ $= \left[\ln\left|x^2 - 3x + 2\right|\right]_3^4 + 3\left[\frac{1}{2\left(\frac{1}{2}\right)}\ln\left|\frac{x-\frac{3}{2}-\frac{1}{2}}{x-\frac{3}{2}+\frac{1}{2}}\right|\right]_3^4$ $= \left[\ln 6 - \ln 2\right] + 3\left[\ln\frac{2}{3} - \ln\frac{1}{2}\right]$ $= \ln\frac{6}{2} + 3\left(\ln 2 - \ln 3 + \ln 2\right)$ $= 6\ln 2 - 2\ln 3$

OR

$$\int_3^4 \frac{2x}{x^2-3x+2}\,\mathrm{d}x$$

$$= \int_3^4 \frac{2x-3}{x^2-3x+2}\,\mathrm{d}x + \int_3^4 \frac{3}{(x-1)(x-2)}\,\mathrm{d}x$$

$$= \Big[\ln\left|x^2-3x+2\right|\Big]_3^4 + 3\int_3^4 \frac{1}{x-2} - \frac{1}{x-1}\,\mathrm{d}x$$

$$= \Big[\ln\left|x^2-3x+2\right|\Big]_3^4 + 3\Big[\ln|x-2| - \ln|x-1|\Big]_3^4$$

$$= \ln 6 - \ln 2 + 3(\ln 2 - \ln 3 - \ln 1 + \ln 2)$$

$$= 6\ln 2 - 2\ln 3$$

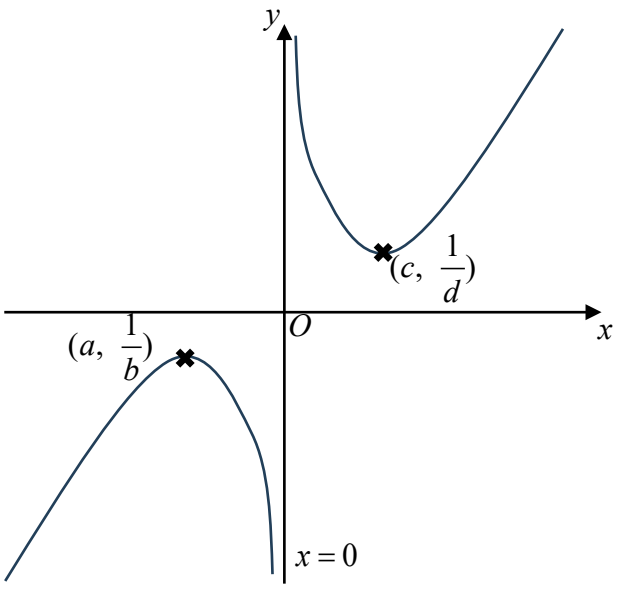
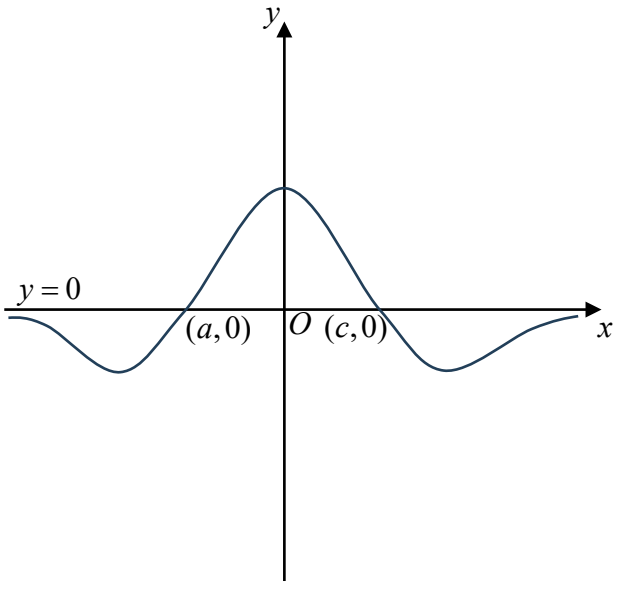
$$\therefore \int_3^4 \left(\frac{12\ln x}{x^2} + \frac{2x}{x^2-3x+2}\right)\mathrm{d}x$$

$$= (1 - 6\ln 2 + 4\ln 3) + (6\ln 2 - 2\ln 3)$$

$$= 1 + 2\ln 3$$

$$= 1 + \ln 9 \quad \text{where } C = 1 \text{ and } D = 9$$

Qn	Suggested Solutions
5(a)	$k = -\dfrac{b}{10}$ The function would be **undefined** at $x = k = -\dfrac{b}{10}$.
5(b)	Let $y = \dfrac{a-x}{10x+b}$ $10xy + by = a - x$ $10xy + x = a - by$ $x = \dfrac{a-by}{10y+1}$ $\mathrm{f}^{-1}(x) = \dfrac{a-bx}{10x+1}$ $\mathrm{R}_{\mathrm{f}^{-1}} = \mathbb{R} \setminus \left\{-\dfrac{b}{10}\right\}$ (or equivalent)
5(c)	Given that $\mathrm{f}(x) = \mathrm{f}^{-1}(x)$ for all x in the domain of f $\Rightarrow \dfrac{a-x}{10x+b} = \dfrac{a-bx}{10x+1}$ Hence, by observation, $\underline{b = 1.}$ For the inverse function to exist, note that f must be a one-one function. Notice that: $\mathrm{f}(x) = \dfrac{a-x}{10x+1} = \dfrac{a-x}{-10\left(-\dfrac{1}{10}-x\right)}$ $\Rightarrow$ if $a = -\dfrac{1}{10}$, $\mathrm{f}(x) = \dfrac{-\dfrac{1}{10}-x}{-10\left(-\dfrac{1}{10}-x\right)} = -\dfrac{1}{10}$ which is a constant function, for which f^{-1} does not exist $\therefore a \neq -\dfrac{1}{10}$ $a \in \mathbb{R} \setminus \left\{-\dfrac{1}{10}\right\}$ or equivalent
5(d)	$\underline{\dfrac{2(a-b)}{11}}$

Qn	Suggested Solutions
6(a)	1. Translate 1 unit in the negative x-direction 2. Scale by a factor of $\sqrt{3}$ parallel to the x-axis **OR** 1. Scale by a factor of $\sqrt{3}$ parallel to the x-axis 2. Translate $\sqrt{3}$ units in the negative x-direction
6(b)(i)	
6(b)(ii)	

Qn	Suggested Solutions
7(a)	$\frac{x^2}{4k^2}+\frac{y^2}{16k^4}=1$
7(b)	$\frac{x^2}{4k^2}+\frac{y^2}{16k^4}=1$ $4k^2x^2+y^2=16k^4$ $y^2=16k^4-4k^2x^2$ $y^2=4k^2\left(4k^2-x^2\right)$ $y=\sqrt{4k^2\left(4k^2-x^2\right)}$ $y=2k\sqrt{4k^2-x^2}$ $A=xy$ $=x\left(2k\sqrt{4k^2-x^2}\right)$ $=2kx\sqrt{4k^2-x^2}$
	$A^2=4k^2x^2\left(4k^2-x^2\right)$ $2A\frac{\mathrm{d}A}{\mathrm{d}x}=32k^4x-16k^2x^3$ Let $\frac{\mathrm{d}A}{\mathrm{d}x}=0 \Rightarrow 16k^2x\left(2k^2-x^2\right)=0$ $\Rightarrow 16k^2x\left(\sqrt{2}k-x\right)\left(\sqrt{2}k+x\right)=0$ $x=0$ (reject) or $x=\sqrt{2}k$ or $x=-\sqrt{2}k$ (reject) By first derivative test,

	$x=\sqrt{2}k^-$	$x=\sqrt{2}k$	$x=\sqrt{2}k^+$
$\sqrt{2}k-x$	+ve	0	–ve
$\frac{\mathrm{d}A}{\mathrm{d}x}$	+ve	0	–ve
Slope			

Area is maximum when $x=\sqrt{2}k$

Maximum area of the rectangular patio $OMPN$ $A=2k\sqrt{2}k\sqrt{4k^2-2k^2}=4k^3$ m^2.

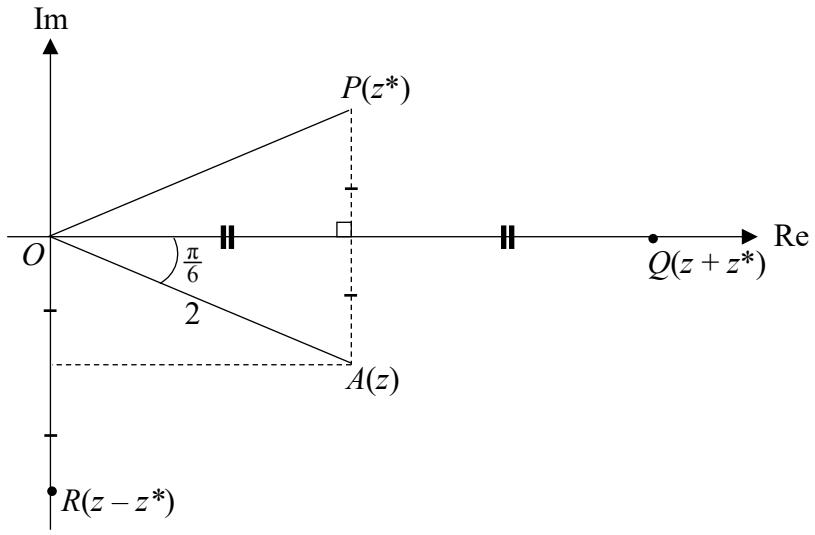
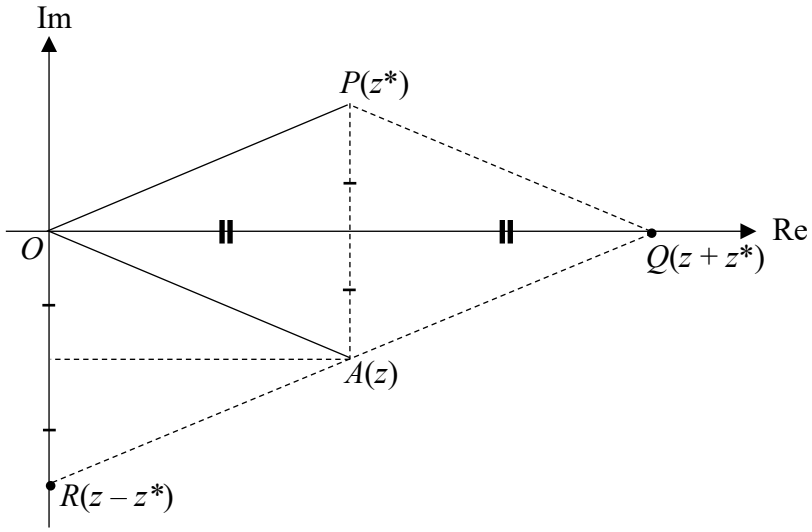
Alternative method

By second derivative test,

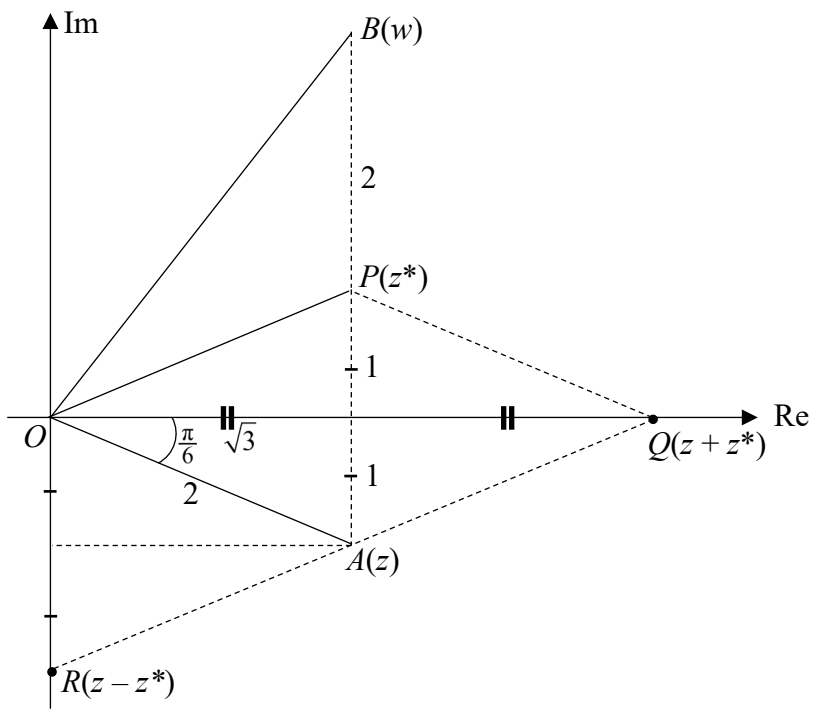
$2\left(\frac{\mathrm{d}A}{\mathrm{d}x}\right)^2+2A\frac{\mathrm{d}^2A}{\mathrm{d}x^2}=32k^4-48k^2x^2$

when $x=\sqrt{2}k$,

	$2(0)^2 + 2\left(4k^3\right)\dfrac{\mathrm{d}^2A}{\mathrm{d}x^2} = 32k^4 - 48k^2\left(2k^2\right)$ $\dfrac{\mathrm{d}^2A}{\mathrm{d}x^2} = -8k < 0$ Area is maximum when $x = \sqrt{2}k$ Maximum area of the rectangular patio $OMPN$ $A = 2k\sqrt{2}k\sqrt{4k^2 - 2k^2} = 4k^3$ m^2.
7(c)	Total area of the garden plot $OXY = \dfrac{1}{4}\pi(2k)\left(4k^2\right)$ $= 2\pi k^3$ m^2 Maximum Ratio = $\dfrac{4k^3}{2\pi k^3} = \dfrac{2}{\pi}$ which is a constant value, independent of k.

Qn	Suggested Solutions
8(a)	
8(b)	 $OPQR$ is a trapezium A lies on line segment QR as its coordinates are (Re(z), Im(z)) which is exactly the midpoint of R(0, 2Im(z)) and Q(2Re(z), 0)

8(c)



Note: Since

$z - w^* = 2\text{i}$

$w^* = z - 2\text{i}$

$w = z^* + 2\text{i}$

i.e. B should be (2 units) vertically above P

Since $\arg(z) = -\dfrac{\pi}{6}$ and $|z| = 2$,

$\text{Re}(z) = \sqrt{3}$ and $\text{Im}(z) = -1$

Hence:

$\text{Im}(w) = 3$

$\arg(w) = \tan^{-1}\dfrac{3}{\sqrt{3}}$

$= \dfrac{\pi}{3}$

Otherwise:

$z = \sqrt{3} - \text{i}$

$z - w^* = 2\text{i}$

$w^* = z - 2\text{i}$

$= \sqrt{3} - \text{i} - 2\text{i}$

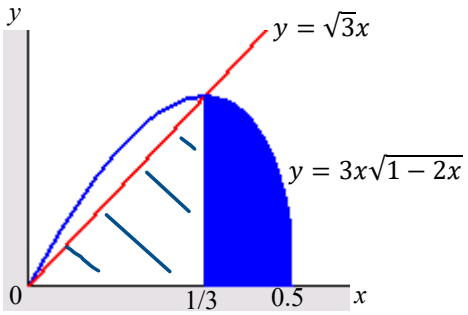
$= \sqrt{3} - 3\text{i}$

$w = \sqrt{3} + 3\text{i}$

	$\arg(w) = \tan^{-1}\dfrac{3}{\sqrt{3}}$ $= \dfrac{\pi}{3}$

Qn	Suggested Solutions
9(a)	$y = \sqrt{2+\cos x}$ $\Rightarrow \quad y^2 = 2+\cos x \quad$ --- (1) Differentiating w.r.t. x, $2y\dfrac{dy}{dx} = -\sin x$ Differentiating w.r.t. x, $2\left(\dfrac{dy}{dx}\right)^2 + 2y\dfrac{d^2y}{dx^2} = -\cos x$ $2\left(\dfrac{dy}{dx}\right)^2 + 2y\dfrac{d^2y}{dx^2} = -\left(y^2-2\right)$ using (1) $2y\dfrac{d^2y}{dx^2} + 2\left(\dfrac{dy}{dx}\right)^2 + y^2 = 2$ (shown)
9(b)	From (a), $2y\dfrac{d^2y}{dx^2} + 2\left(\dfrac{dy}{dx}\right)^2 + y^2 = 2$ Differentiating w.r.t. x, $2\left(\dfrac{dy}{dx}\right)\dfrac{d^2y}{dx^2} + 2y\dfrac{d^3y}{dx^3} + 4\left(\dfrac{dy}{dx}\right)\dfrac{d^2y}{dx^2} + 2y\dfrac{dy}{dx} = 0$ $6\left(\dfrac{dy}{dx}\right)\dfrac{d^2y}{dx^2} + 2y\dfrac{d^3y}{dx^3} + 2y\dfrac{dy}{dx} = 0$ Differentiating w.r.t. x, $6\left(\dfrac{d^2y}{dx^2}\right)^2 + 6\left(\dfrac{dy}{dx}\right)\dfrac{d^3y}{dx^3} + 2\left(\dfrac{dy}{dx}\right)\dfrac{d^3y}{dx^3} + 2y\dfrac{d^4y}{dx^4} + 2\left(\dfrac{dy}{dx}\right)^2 + 2y\dfrac{d^2y}{dx^2} = 0$ $6\left(\dfrac{d^2y}{dx^2}\right)^2 + 8\left(\dfrac{dy}{dx}\right)\dfrac{d^3y}{dx^3} + 2y\dfrac{d^4y}{dx^4} + 2\left(\dfrac{dy}{dx}\right)^2 + 2y\dfrac{d^2y}{dx^2} = 0$ When $x = 0$, $y = \sqrt{2+\cos 0} = \sqrt{3}$ $2(\sqrt{3})\dfrac{dy}{dx} = -\sin 0 \Rightarrow \dfrac{dy}{dx} = 0$ $2(\sqrt{3})\dfrac{d^2y}{dx^2} + 2(0)^2 + \left(\sqrt{3}\right)^2 = 2 \Rightarrow \dfrac{d^2y}{dx^2} = -\dfrac{1}{2\sqrt{3}}$ $6(0)\left(-\dfrac{1}{2\sqrt{3}}\right) + 2(\sqrt{3})\dfrac{d^3y}{dx^3} + 2(\sqrt{3})(0) = 0 \Rightarrow \dfrac{d^3y}{dx^3} = 0$ $6\left(-\dfrac{1}{2\sqrt{3}}\right)^2 + 8(0)(0) + 2\sqrt{3}\dfrac{d^4y}{dx^4} + 2(0)^2 + 2\sqrt{3}\left(-\dfrac{1}{2\sqrt{3}}\right) = 0$

	$\frac{1}{2}+2\sqrt{3}\frac{d^4y}{dx^4}-1=0 \Rightarrow \frac{d^4y}{dx^4}=\frac{1}{4\sqrt{3}}$ Thus Maclaurin expansion is $y=\sqrt{2+\cos x}=\sqrt{3}+0x-\frac{1}{2\sqrt{3}}\left(\frac{x^2}{2!}\right)+0\left(\frac{x^3}{3!}\right)+\frac{1}{4\sqrt{3}}\left(\frac{x^4}{4!}\right)$ $=\sqrt{3}-\frac{1}{4\sqrt{3}}x^2+\frac{1}{96\sqrt{3}}x^4+.....$ $=\sqrt{3}-\frac{\sqrt{3}}{12}x^2+\frac{\sqrt{3}}{288}x^4+.....$
9c	$\sqrt{2+\cos x}=(2+\cos x)^{\frac{1}{2}}$ $=\left(2+\left(1-\frac{x^2}{2}+\frac{x^4}{24}-.....\right)\right)^{\frac{1}{2}}$ $=\sqrt{3}\left(1+\left(-\frac{x^2}{6}+\frac{x^4}{72}+.....\right)\right)^{\frac{1}{2}}$ $=\sqrt{3}\left(1+\frac{1}{2}\left(-\frac{x^2}{6}+\frac{x^4}{72}+...\right)+\frac{\frac{1}{2}\left(-\frac{1}{2}\right)}{2!}\left(-\frac{x^2}{6}+\frac{x^4}{72}+...\right)^2+...\right)$ $=\sqrt{3}\left(1-\frac{x^2}{12}+\frac{x^4}{144}-\frac{1}{8}\left(\frac{x^4}{36}+...\right)+...\right)$ $=\sqrt{3}-\frac{\sqrt{3}}{12}x^2+\frac{\sqrt{3}}{288}x^4+....$ (verified same as part (b))

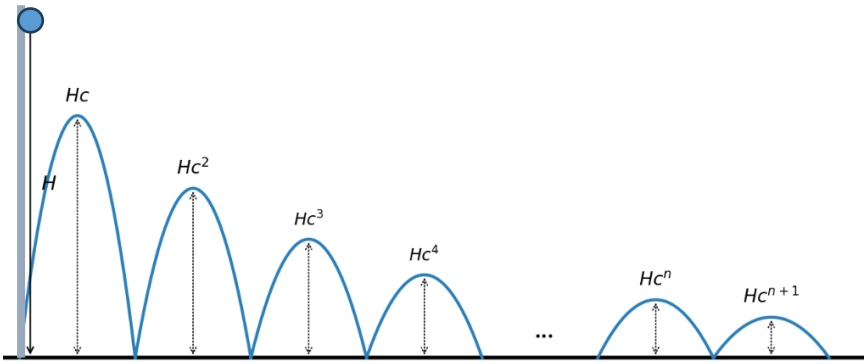
Qn	Suggested Solutions
10(a)	Required Volume $= \pi \int_0^{0.5} y^2 \, dx$ $= \pi \int_0^{0.5} (9x^2 - 18x^3) \, dx$ $= \pi \left[3x^3 - 4.5x^4 \right]_0^{0.5}$ $= 0.09375\pi$ or $\frac{3\pi}{32}$ unit^3
10(b)	$\int_{1/3}^{1/2} 3x\sqrt{1-2x} \, dx$ $= \int_{1/\sqrt{3}}^{0} \frac{3}{2}(1-v^2)(v)(-v) \, dv$ $= \frac{3}{2} \int_{1/\sqrt{3}}^{0} (v^4 - v^2) \, dv$ $= \frac{3}{2} \left[\frac{v^5}{5} - \frac{v^3}{3} \right]_{1/\sqrt{3}}^{0}$ $= \frac{3}{2} \left(\frac{(1/\sqrt{3})^3}{3} - \frac{(1/\sqrt{3})^5}{5} \right)$ $= \frac{1}{2}\left(\frac{1}{3}\right)^{1.5} - \frac{3}{10}\left(\frac{1}{3}\right)\left(\frac{1}{3}\right)^{1.5} = \frac{2}{5}\left(\frac{1}{3}\right)^{1.5}$ where $k = \frac{2}{5}$
10(c)	 When $\sqrt{3}x = 3x\sqrt{1-2x}$, $x = 0$ or $x = \frac{1}{3}$ Hence the required area $= \frac{1}{2}\left(\frac{1}{3}\right)\left(\frac{\sqrt{3}}{3}\right) + \frac{2}{5}\left(\frac{1}{3}\right)^{1.5}$ $= \frac{1}{2}\left(\frac{1}{3}\right)\left(\frac{\sqrt{3}}{3}\right) + \frac{2}{5}\left(\frac{1}{3\sqrt{3}}\right)$ $= \frac{\sqrt{3}}{10}$

Qn	Suggested Solutions
11(a)	Normal of the plane $ABC = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} -2 \\ 0 \\ 4 \end{pmatrix} \times \begin{pmatrix} 1 \\ -1 \\ 5 \end{pmatrix} = \begin{pmatrix} 4 \\ 14 \\ 2 \end{pmatrix} = 2\begin{pmatrix} 2 \\ 7 \\ 1 \end{pmatrix}$ $\Pi_{ABC} : \mathbf{r} \cdot \begin{pmatrix} 2 \\ 7 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 7 \\ 1 \end{pmatrix} = 15$ $\Rightarrow 2x + 7y + z = 15$
11(b)	$\begin{pmatrix} p \\ 0 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 7 \\ 1 \end{pmatrix} = 0$ $\Rightarrow p = -1$
11(c)	$L : \underset{\sim}{r} = \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 7 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}$ When L and plane ABC intersect, $\begin{pmatrix} -1+2\lambda \\ 7\lambda \\ 2+\lambda \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 7 \\ 1 \end{pmatrix} = 15$ $\Rightarrow -2 + 4\lambda + 49\lambda + 2 + \lambda = 15$ $\Rightarrow \lambda = \frac{5}{18}$ $\overrightarrow{OF} = \begin{pmatrix} -1+\frac{5}{9} \\ \frac{35}{18} \\ 2+\frac{5}{18} \end{pmatrix} = \begin{pmatrix} -\frac{4}{9} \\ \frac{35}{18} \\ \frac{41}{18} \end{pmatrix}$ Coordinates of $F\left(-\frac{4}{9}, \frac{35}{18}, \frac{41}{18}\right)$

11(d)	Let the position vector of the reflection of the point D be $\overrightarrow{OD'}$ $\overrightarrow{OD'} = 2\overrightarrow{OF} - \overrightarrow{OD}$ $= 2\begin{pmatrix} -\frac{4}{9} \\ \frac{35}{18} \\ \frac{41}{18} \end{pmatrix} - \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} \frac{1}{9} \\ \frac{35}{9} \\ \frac{23}{9} \end{pmatrix}$

Qn | **Suggested Solutions**

12



1st Impact | 2nd Impact | 3rd Impact | 4th Impact | n-th Impact | $(n+1)$-th Impact

r	Height ball rebounds to **after** rth impact
1	Hc^1
2	Hc^2
3	Hc^3
…	
n	Hc^n

12(a)

Height the ball rebounds to right after the 3rd impact = Hc^3

12(b)

D_n = Total vertical distance travelled by the ball from the instant it is first dropped until just before it impacts the floor for the $(n+1)$th time = height of first drop H + two times subsequent rebound heights (since ball must go up and come back down after first drop)

$$= H + 2\left(Hc^1 + Hc^2 + Hc^3 + ... + Hc^n\right)$$

$$= H + 2\left(\frac{Hc(1-c^n)}{1-c}\right)$$

$$= H\left(1 + \frac{2c(1-c^n)}{1-c}\right)$$

$$= H\left(\frac{1-c+2c-2c^{n+1}}{1-c}\right)$$

$$= H\left(\frac{1+c-2c^{n+1}}{1-c}\right) \quad \text{(shown)}$$

12(c)(i)

Using H=5 and c = 0.9.

$$D_n = 5\left(\frac{1+0.9-2(0.9)^{n+1}}{1-0.9}\right) = 50\left(1.9-2(0.9)^{n+1}\right)$$

Using GC Table:

n	Y= $D_n = 50\left(1.9-2(0.9)^{n+1}\right)$	Interpretation: Total vertical distance travelled after nth impact and just before just before $(n+1)$th impact
1	14	Total vertical distance travelled after 1st impact and just before 2nd impact = 14
…	…	

10	63.6189	Total vertical distance travelled after 10th impact and just before 11th impact = 63.6 < 65 (not yet) $\Rightarrow$ need to go on to 11th impact
11	66.757	Total vertical distance travelled after 11th impact and just before 12th impact = 66.8 > 65 $\Rightarrow$ happens before 12th impact

Since 63.619 < 65 < 66.757, the ball reaches a total vertical distance travelled of 65 metres strictly after its 11th floor impact, but prior to its 12th floor impact.
$\Rightarrow$ the ball has impacted the floor **11** times.

Algebraic Method:

Solving $D_n = 50\left(1.9 - 2(0.9)^{n+1}\right) \geq 65$

$50\left(1.9 - 2(0.9)^{n+1}\right) \geq 65$

$(0.9)^{n+1} \leq \frac{1}{-2}\left(\frac{65}{50} - 1.9\right)$

$\Rightarrow (0.9)^{n+1} \leq 0.3$

$\Rightarrow n \geq \frac{\ln 0.3}{\ln 0.9} - 1 \approx 10.427$

*can use GC graph/table to solve for *n* when using algebraic method also

Ball would have travelled 65metres just before ("10.427"+1)-th impact ≈ 12 (rounded up))th impact

$\Rightarrow$ the ball has impacted the floor **11** times.

12(c)(ii)

Happens between 11th and 12th impact

$\Rightarrow$ rebound height after 11th impact $= (5)(0.9)^{11} = 1.56903$

(or rebound height after 11th impact

$= \frac{D_{11} - D_{10}}{2} = \frac{66.757 - 63.619}{2} = 1.569$)

Vertical distance needed to reach exactly 65m after 11th impact

$65 - 63.6189 = 1.3811 < 1.569$ (rebound height after 11th impact)

Thus, ball is still travelling upwards and the height of the ball from the floor is 1.38m (to 3sf)

Alternative:

Distance ball would have travelled at maximum height reached after 11th impact

$= 63.619 + 1.569 = 65.188 > 65$

Note: To explain why the maximum height does not correspond to "*n* = 10.5":

At maximum height, total vertical distance travelled

$= 63.619 + 1.569 = 65.188$ (rounded off answers from above)

$(= 50\left(1.9 - 2(0.9)^{10+1}\right) + (5)(0.9)^{11} = 65.18799337)$ (more accurate)

Solving $D_n = 65.18799337 = 50\left(1.9 - 2(0.9)^{n+1}\right)$,

$n = 10.48683602$ (not 10.5)

Or when $n = 10.5$,

	$D_n = 50\left(1.9 - 2(0.9)^{10.5+1}\right) = 65.22931287$ (not 65.18799) Note that the formula derived for D_n is only valid for positive integer values of *n,* so by right, we should not be using the formula for non-integer values of *n*.
12(c)(iii)	For the ball in the laboratory test, $D_n = 50\left(1.9 - 2(0.9)^{n+1}\right)$ As $n \to \infty$, $(0.9)^{n+1} \to 0$ $\Rightarrow D_n \to 50\left(1.9 - 2(0)\right) = 95$ Thus, there is a theoretical limit to the total vertical distance that can be travelled by the ball in the laboratory test and this limit is 95m.